

Combined Families Summary

LLM · ASR · TTS bake-off — cross-family overview

LLM (speed)

Nile-Chat-4B

LLM (quality)

Qwen3-8B

ASR pick

Whisper-Large-v3-Turbo-CT2

TTS pick

VoiceTut-TTS

Overview

This summary helps select one model from each family for the Arabic voice robot stack.
All three families now have successful bake-off results and shortlists.
Suggested ASR default: Whisper-Large-v3-Turbo-CT2. Accuracy alternative: Whisper-Large-v3-CT2.
Suggested LLM speed default: Nile-Chat-4B. Quality alternative: Qwen3-8B.
Suggested TTS default (speed/resources): VoiceTut-TTS - confirm by listening to the WAV.
Prefer LOWER latency/memory/error rates; prefer HIGHER accuracy/throughput/quality/robot scores.

How to read this report

Page 2: shared metrics glossary (RTF, TTFT, WER, VRAM, ...).
Page 3: shortlist table and decision path.
Page 4: family readiness and top scores.
Pages 5-7: LLM, ASR, and TTS snapshot charts.
Page 8: package contents and bottom line.

Combined - Metrics glossary (what to prefer)

Use this page while reading the charts. Direction labels show what you want for a production robot.

TTFT - Time To First Token (s)

LOWER better

Delay until the model starts generating. Lower = snappier turn-taking (robot starts sooner).

Throughput (tokens/s)

HIGHER better

How fast tokens are generated after the first one. Higher = long answers finish sooner.

Generate time (s)

LOWER better

Total time to finish the reply. Lower is better (also depends on answer length).

Accuracy % (word accuracy)

HIGHER better

Share of words recognized correctly (about 1 - WER). Higher = fewer recognition mistakes.

WER - Word Error Rate

LOWER better

Fraction of words wrong. 0 = perfect. Lower = better recognition.

CER - Character Error Rate

LOWER better

Same idea as WER at character level. Useful for Arabic. Lower = better.

Chars/s (characters per second)

HIGHER better

Input characters synthesized per second of compute. Higher = faster speech generation.

Peak VRAM (MB)

LOWER better

GPU memory used at peak. Lower leaves room to run ASR + LLM + TTS together on one GPU.

Peak RAM (MB)

LOWER better

System (CPU) memory used at peak. Lower is safer on smaller servers / VPS hosts.

Peak CPU %

LOWER better

CPU utilization spike during the run. Lower means less contention with other services.

Cold load time (s)

LOWER better

Seconds to load the model into memory the first time. Matters for startup / cold starts.

RTF (Real-Time Factor)

LOWER better

Processing time / audio duration. RTF < 1 = faster than realtime (good). RTF > 1 = slower than the audio.

x Realtime (xRT)

HIGHER better

How many times faster than realtime (about 1/RTF). Example: 6x = 6s of audio in ~1s of compute.

Robot realtime score (0-100)

HIGHER better

Composite score for robot use (speed + resources + accuracy where relevant). Higher = better fit.

Remember

Shared rule: LOWER = TTFT, RTF, WER, CER, VRAM, RAM, CPU%, load/generate/transcribe time.
Shared rule: HIGHER = accuracy%, tokens/s, chars/s, quality scores, robot/balanced scores, xRealtime.
See each family PDF glossary for the full term list.

Context only

Duration of the produced WAV. Not good/bad alone — used with generate time to get RTF.

HIGHER better

Share of runs that completed without error. Prefer 100% for production and disk.

Recommended shortlist

ASR — robot default

Whisper-Large-v3-Turbo-CT2

Best accuracy/speed/VRAM blend for realtime input

ASR — accuracy alt

Whisper-Large-v3-CT2

Highest word accuracy when recognition quality is critical

LLM — speed default

Nile-Chat-4B

Best TTFT/throughput/VRAM composite for turn-taking

LLM — quality alt

Qwen3-8B

Best rubric quality (4.65/5) and auto-pass rate

TTS — speed/resources

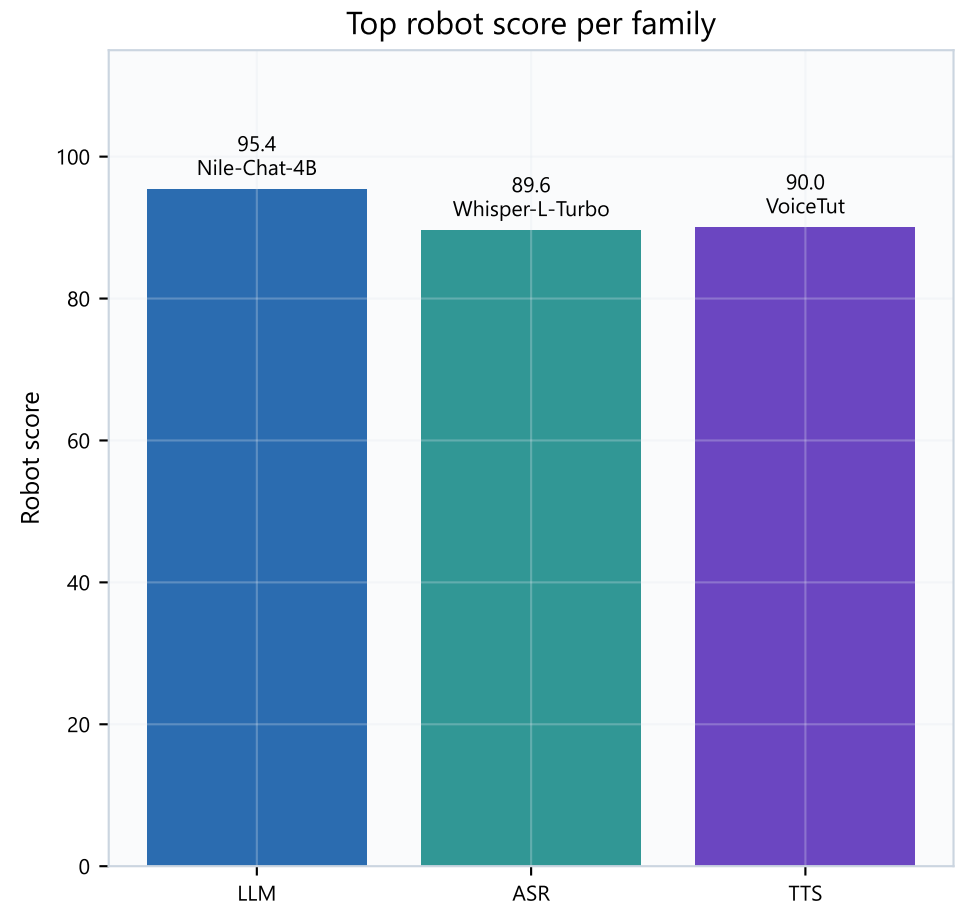
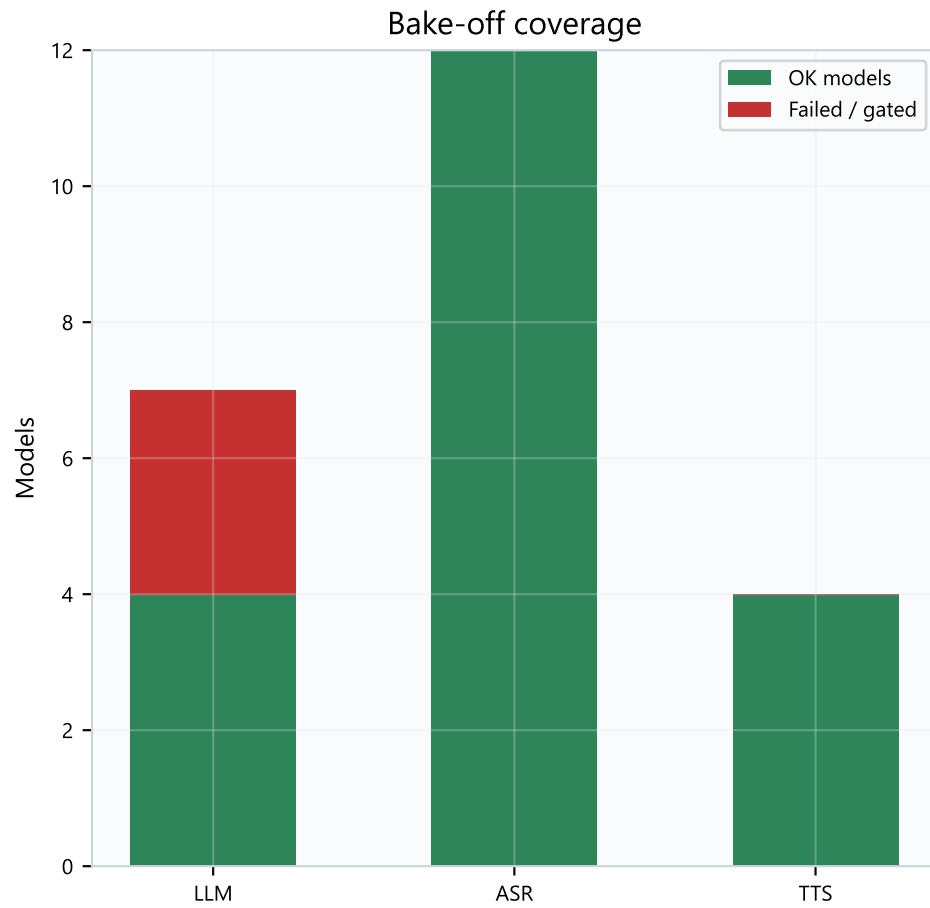
VoiceTut-TTS

Best RTF/throughput/VRAM composite — listen to WAV before freeze

Decision path

- 1) Lock ASR first — input quality gates the whole pipeline.
- 2) Lock LLM next — choose speed (Nile-Chat-4B) or quality (Qwen3-8B / Qwen3-4B).
- 3) Lock TTS after listening — default VoiceTut-TTS; fallback SILMA-TTS if voice quality wins.
- 4) Validate the combined VRAM budget on the deployment GPU.

Family readiness & top robot scores

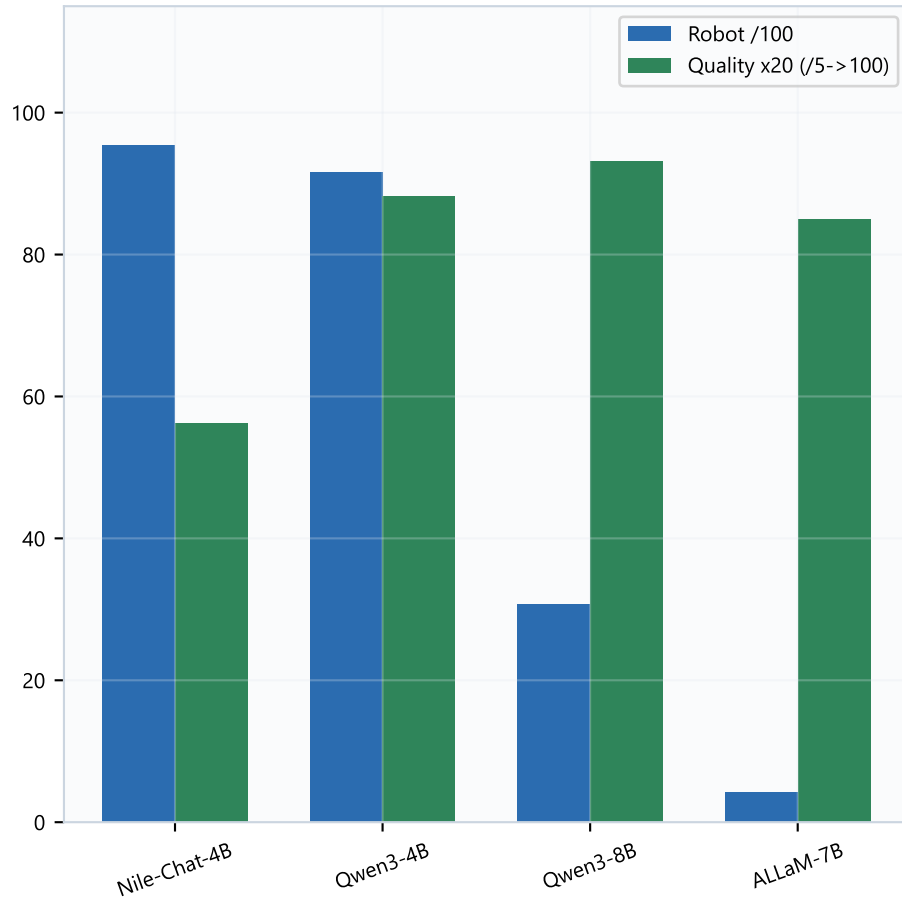


Readiness reading

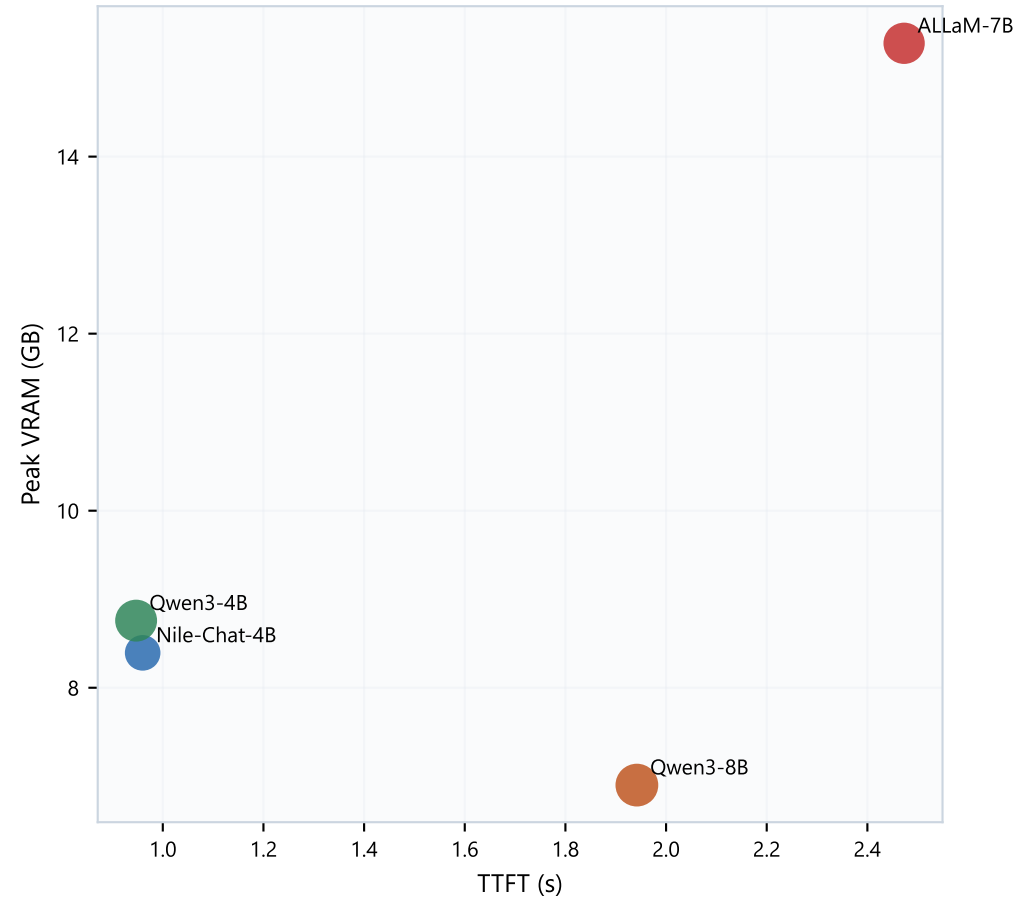
ASR is fully covered (12/12 OK). LLM has 4 OK models + 3 gated failures.
TTS is now ready for shortlisting (4/4 OK) — VoiceTut leads robot score.
All three families can be shortlisted together; only TTS still needs a human listening pass.
Plan GPU headroom for Whisper + LLM + VoiceTut/SILMA together.

LLM snapshot

Speed composite vs quality



Bubble size ~ quality

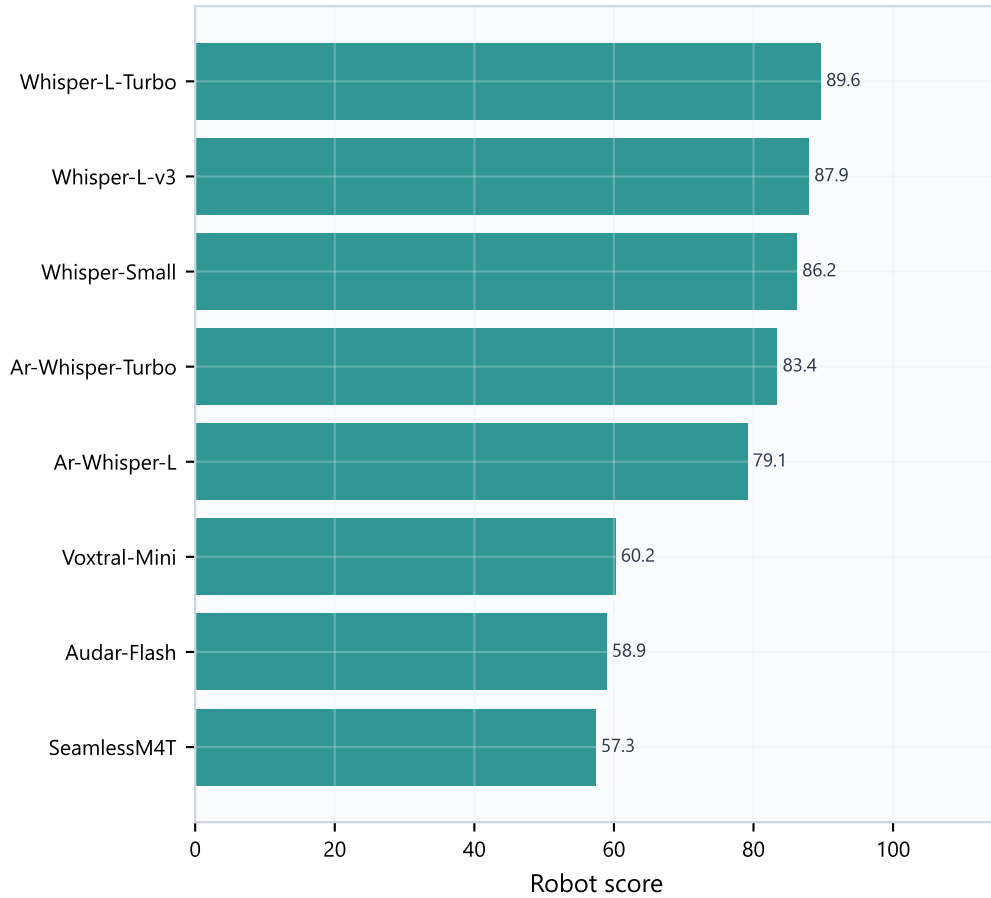


LLM takeaway

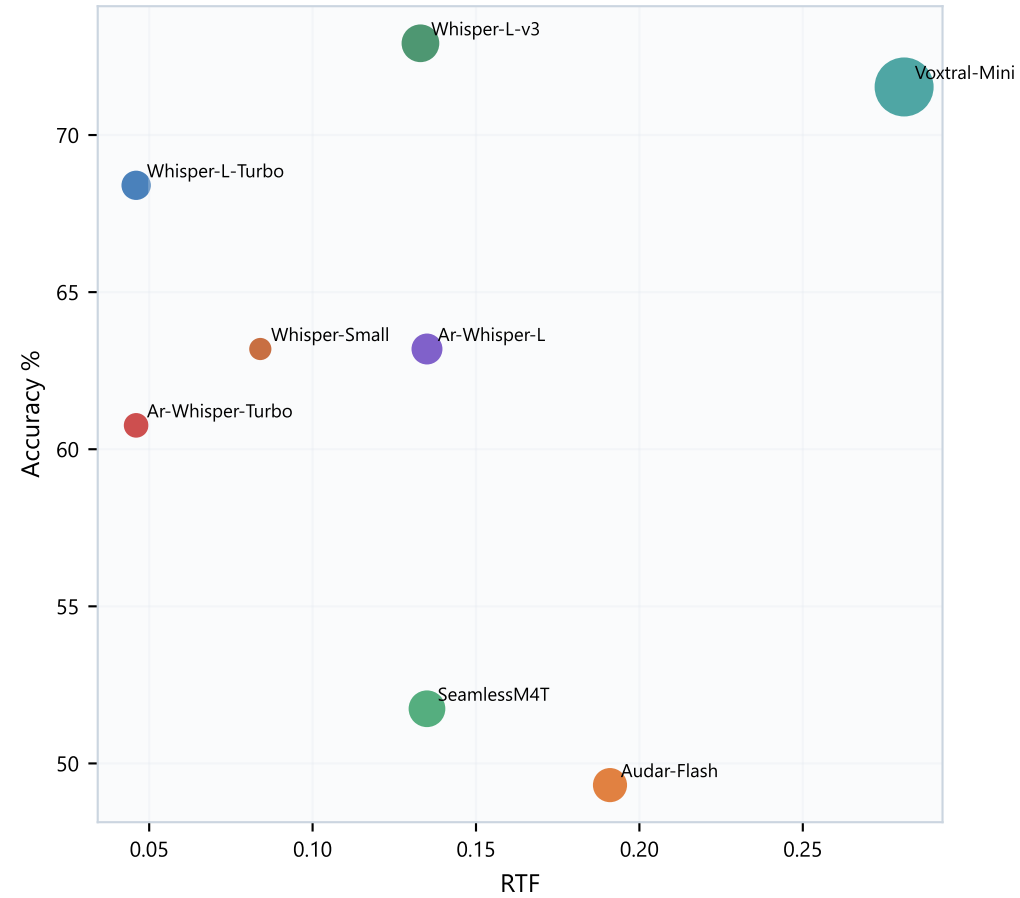
Nile-Chat-4B wins robot realtime; Qwen3-8B wins quality.
If spoken answers must stay short and correct, lean quality (Qwen).
If the robot must start speaking immediately, lean Nile-Chat-4B — then spot-check verbosity.
Qwen3-4B is the practical middle ground for many deployments.

ASR snapshot (top 8)

Robot leaderboard



Accuracy vs speed (bubble = VRAM)

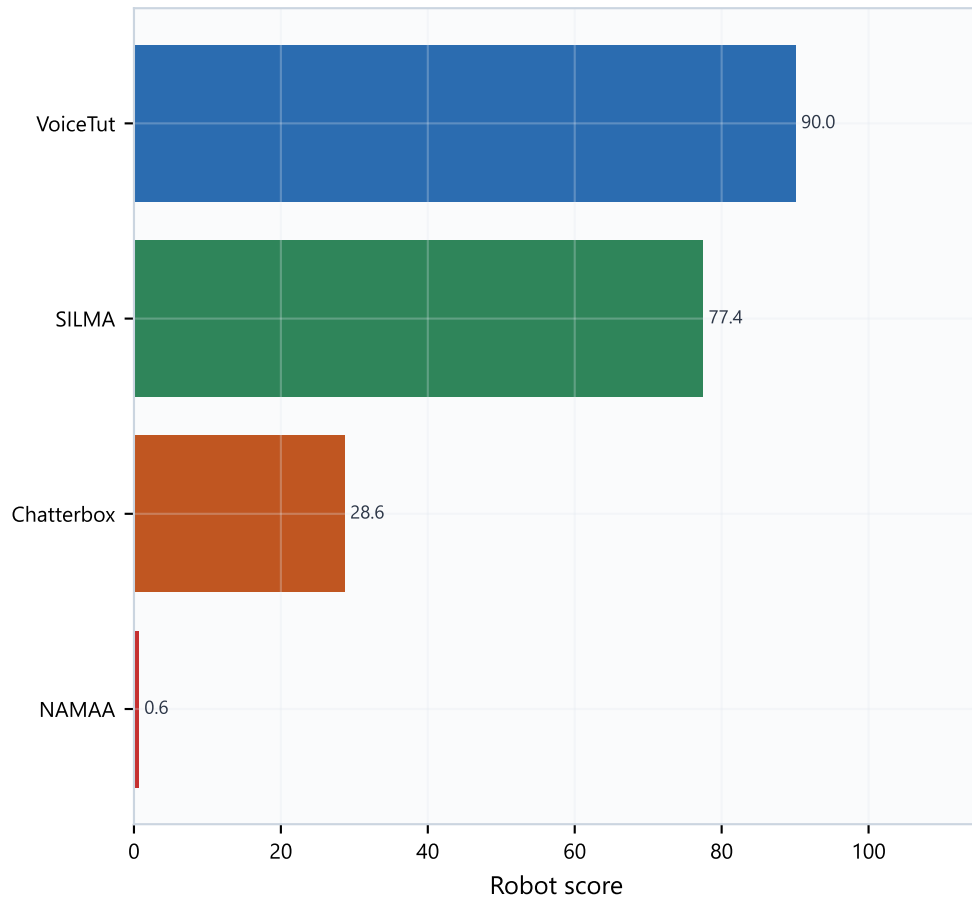


ASR takeaway

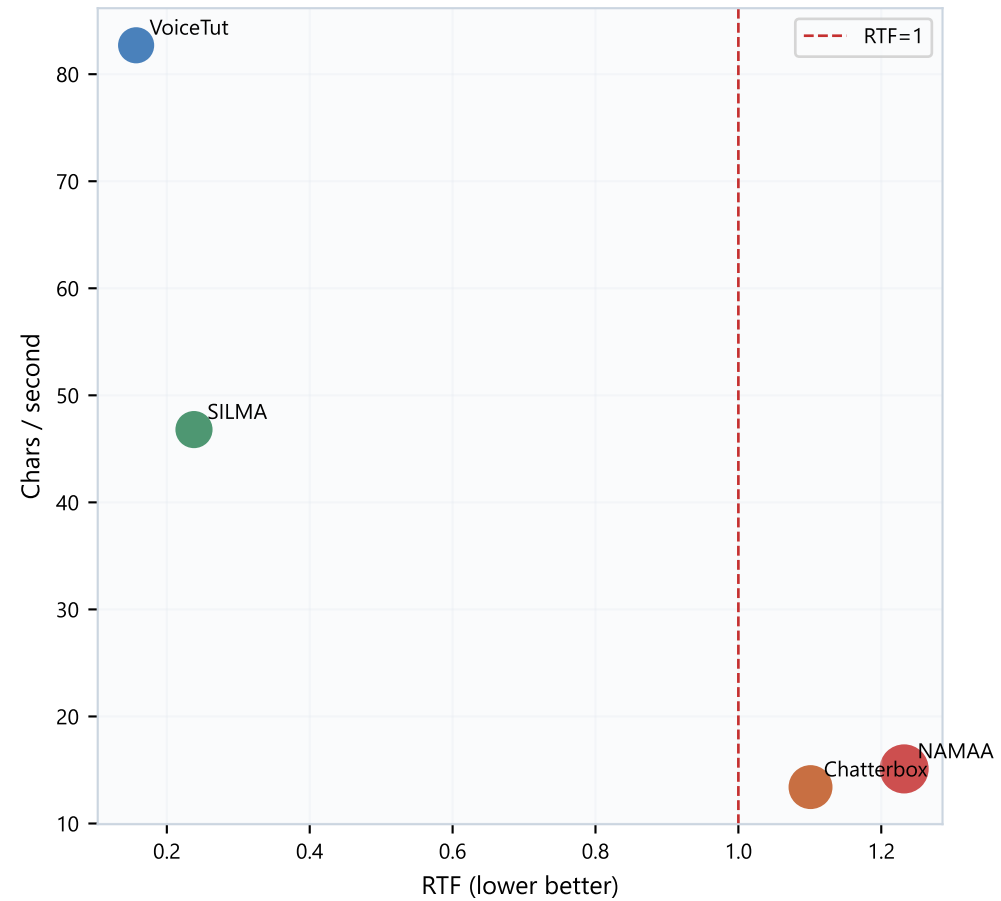
Default: Whisper-Large-v3-Turbo-CT2 for robot balance.
Accuracy alternative: Whisper-Large-v3-CT2.
Whisper CT2 family occupies the useful top band; Qwen ASR variants lag badly on WER.
Keep ASR VRAM low so the LLM + TTS still fit on the same GPU.

TTS snapshot

TTS robot leaderboard



Speed vs throughput (bubble = VRAM)



TTS takeaway

Default speed/resources pick: VoiceTut-TTS.
SILMA-TTS is the best realtime alternative if VoiceTut fails the listen test.
Chatterbox and NAMAA are not realtime on this bake-off — use only if voice quality clearly wins.
Always listen to WAVs before freezing the production voice.

Package contents & bottom line

PDF package

LLM_Model_Selection_Report.pdf — charts + picks + quality guidance
ASR_Model_Selection_Report.pdf — charts + accuracy/speed guidance
TTS_Model_Selection_Report.pdf — charts + picks + listening checklist
Combined_Families_Summary.pdf — this cross-family overview

WAV files for listening: analytics/final/tts_outputs/*.wav

Bottom line

ASR = Whisper-Large-v3-Turbo-CT2 (or Whisper-Large-v3-CT2 for max accuracy).
LLM = Nile-Chat-4B (speed) or Qwen3-8B (quality).
TTS = VoiceTut-TTS pending listening confirmation (SILMA as backup).
See family PDFs for full evidence before final sign-off.